

Machine Learning Algorithm Comparison

COS30083 Applied Machine Learning — Algorithm & Learning Paradigm Reference Guide

1. Main Machine Learning Paradigms Comparison

Paradigm	Input Data & Labels	Core Learning Goal	Key Tasks & Examples
Supervised Learning	Labeled dataset with ground truth targets (X, y)	Learn a mapping function $g \approx f$ from inputs to target outputs to accurately predict unseen data.	• Regression: House price prediction • Classification: Email spam filter • Ranking & Sequence prediction
Unsupervised Learning	Unlabeled data only (X without targets y)	Discover underlying patterns, hidden structures, or natural clusterings within input data.	• Clustering: Customer segmentation • Association Rule Mining • Dimensionality Reduction (PCA)
Reinforcement Learning	Environment state, actions, rewards & penalties	Learn an optimal decision-making policy to maximize cumulative rewards through trial and error over time.	• Decision Making: AlphaGo, Game AI • Autonomous Robot Control • Automated Trading Strategies
Semi-Supervised Learning	Small labeled dataset + large unlabeled dataset	Train an initial model on labeled data, make high-confidence predictions on unlabeled data, and retrain iteratively.	• Medical image analysis with expensive annotation costs • Speech recognition with scarce labeled data
Self-Supervised Learning	Unlabeled data where the model generates pseudo-labels	Predict missing or masked parts of input data using self-generated supervisory signals.	• Masked Language Modeling (e.g., BERT) • Self-supervised vision transformers

Key Takeaway — Choosing the Right Learning Paradigm:

Supervised Learning requires explicit target labels y for every training example. When labels are unavailable or expensive to acquire, **Unsupervised Learning** surfaces natural groupings, while **Semi-Supervised** and **Self-Supervised** methods leverage large unlabeled corpora to boost performance.

2. Core Machine Learning Algorithms Comparison

Algorithm	Paradigm & Task	Target Output (y)	Core Mechanism & Key Equations	Optimization & Loss
Linear Regression	Supervised (Regression)	Continuous real value (e.g., price, height, temperature)	Assumes a linear relationship between features and target: $y = w_0x_0 + \dots + w_nx_n + b = w^Tx + b$	Minimizes Mean Squared Error (MSE): $L = \sum (y_{\text{actual}} - y_{\text{predicted}})^2$ via Normal Equation or Gradient Descent
Logistic Regression	Supervised (Classification)	Discrete probability ($0 \leq y \leq 1$ for binary classification)	Applies Sigmoid activation to a linear combination: $\sigma(z) = 1 / (1 + e^{-z})$ where $z = w^Tx + b$	Minimizes Binary Cross-Entropy Loss (Log Loss) iteratively using Gradient Descent
Decision Tree	Supervised (Classification & Regression)	Discrete class labels or continuous numerical values	Recursively partitions data into subsets using a decision tree structure until leaf nodes reach purity.	Evaluates splits using Information Gain (Entropy) or Gini Impurity (classification) / Variance reduction (regression).
Support Vector Machine (SVM)	Supervised (Classification)	Discrete class labels (e.g., +1 or -1)	Finds an optimal separating hyperplane $w^Tx + b = 0$ that maximizes the margin ($1 / w $) to support vectors.	Minimizes Hinge Loss : $L = \max(0, 1 - y \cdot z)$ with regularizer $\frac{1}{2} w ^2$ and penalty parameter C.
K-Nearest Neighbors (KNN)	Supervised (Classification & Regression)	Discrete class or continuous value	Non-parametric instance-based method; assumes nearby data points in feature space belong to the same class.	No explicit training phase (lazy learning); classification uses majority voting of K neighbors; regression uses average.
K-Means Clustering	Unsupervised (Clustering)	Unlabeled cluster assignments (1, 2, ..., K)	Partitioning algorithm; iteratively updates K randomly initialized cluster centroids to the mean of assigned data points.	Minimizes Within-Cluster Sum of Squares (WCSS / Inertia) until cluster assignments stabilize.
Neural Networks	Supervised (Deep Learning)	Discrete categories or continuous values	Multiple layers of interconnected nodes performing matrix multiplications and non-linear activation functions: $y = \text{Softmax}(w \cdot x + b)$	Forward pass predicts outputs → Loss computed → Backpropagation computes gradients → Gradient Descent updates weights.

Quick Revision Rules & Concept Associations:

- 1. **Target Output Rule:** Continuous real values (price, height) → *Regression*; Discrete labels (categories, flags) → *Classification*.
- 2. **SVM Key Terms:** Hyperplane, Support Vectors, Maximum Margin, Hinge Loss, Parameter C.
- 3. **Logistic Regression:** Linear Decision Boundary + Sigmoid Activation (maps score z to 0..1 probability).
- 4. **Neural Networks Pipeline:** Forward Pass → Loss Calculation → Backpropagation → Weight Update via Gradient Descent.